Order Matters

Numerical Expressions

15 marks | 25 minutes | Part A: 10 × 1 mark · Part B: 5 marks

Name _____ Class _____ Score _____ / 15

PART A · CHOOSE THE CORRECT ANSWER

1 Which is a numerical expression for *seven times fifteen minus two times nine*?

A $(7+15)-2\cdot9$ B $7(15-2)\cdot9$
C $7\cdot15-2\cdot9$ D $7\cdot(15-2\cdot9)$

2 Which expression changes 37 °C to °F? (Multiply by 9, divide by 5, add 32.)

A $\frac{37\cdot9}{5}+32$ B $\frac{37\cdot5}{9}+32$
C $\frac{37\cdot9}{5+32}$ D $\frac{37\cdot9}{5}\cdot32$

3 In 9^4, the number 4 is the...

A base B product
C power D exponent

4 Write a numerical expression for: *twelve plus twelve divided by five.*

A $5\div(12+12)$ B $12+12\div5$
C $(12+12)\div5$ D $12\div12+5$

5 To change °C to °F: multiply by 9, divide by 5, and add 32. Which expression changes 20°C to °F?

A $\frac{20\cdot9}{5}+32$ B $\frac{20\cdot5}{9}+32$
C $\frac{20\cdot9}{5+32}$ D $\frac{20\cdot9}{5}\cdot32$

6 Evaluate 6^4.

A 1297 B 1296
C 1295 D 1298

7 Evaluate $72\div8+3\cdot9$.

A 36 B 35
C 38 D 37

8 Evaluate $\frac{(6+6)^2}{2(5-3)}$.

A 37 B 36
C 35 D 38

9 A plumber charges a flat fee of AED 60 plus AED 20 per hour. He works 3 hours on Monday and 3 hours on Tuesday. How much is the bill in AED?

A 180 B 181
C 179 D 182

10 Write a numerical expression for: *two plus five divided by seven.*

A $(2+5)\div7$ B $2+5\div7$
C $7\div(2+5)$ D $2\div5+7$

PART B · SHOW YOUR WORK

11 Lift tickets cost AED 42 for an adult and AED 34 for a child. Ski rental is AED 32 per pair and hot chocolate is AED 4 a cup. A family of 2 adults and 3 children buys tickets, rents 2 pairs of skis and buys one hot chocolate each. (a) Write a numerical expression for the total. (b) Find the total.

Let x Talk

Algebraic Expressions

15 marks | 25 minutes | Part A: 10 × 1 mark · Part B: 5 marks

Name _____ Class _____ Score _____ / 15

PART A · CHOOSE THE CORRECT ANSWER

1 Which verbal expression represents $\frac{2}{5}m^2$?

A two fifths times m squared
B the quotient of two and five times the product of m and two
C two fifths times the product of m and two
D two fifths times m cubed

2 Which is an algebraic expression for *6 more than twice m*?

A 2(m + 6)
B 2m − 6
C 2m + 6
D 6m + 2

3 A gym costs a one-time fee of AED 100 plus AED 30 per month. Which expression gives the cost after **m** months?

A 130m
B 100 + 30m
C 100m + 30m
D 30 + 100m

4 Evaluate $\frac{2b + 3c^2}{4a^2 - 2b}$ if **a = 8, b = 4, c = 16**.

A $\frac{776}{256}$
B $\frac{97}{32}$
C $\frac{97}{31}$
D 3

5 Which verbal expression represents **8(y − 6)**?

A eight times y minus six
B six less than eight times y
C eight times the difference of y and six
D the difference of eight times y and six

6 Write an algebraic expression for: *eight times the quantity n plus 29*.

A 8n + 29
B 8 + n + 29
C 8(n) + 8
D 8(n + 29)

7 A one-day pass costs AED 120 and a three-day pass costs AED 350. Which expression gives the total sales for **n** one-day passes and **t** three-day passes?

A 120 + n + 350 + t
B 120n + 350t
C (120 + 350)(n + t)
D 120t + 350n

8 Evaluate **2(x + z) − y** if **x = 6, y = 20, z = 9**.

A 11
B 10
C 12
D 9

9 A taxi charges a base fee of AED 6 plus AED 3 per kilometre. Write an expression for **k** kilometres and find the cost of a 8 km ride.

A 29
B 31
C 32
D 30

10 Which verbal expression represents **3(x − 7)**?

A the difference of three times x and seven
B seven less than three times x
C three times x minus seven
D three times the difference of x and seven

PART B · SHOW YOUR WORK

11 A taxi charges a AED 1.75 base fee plus AED 3.45 per mile. (a) Write an expression for **m** miles. (b) Find the cost of an 8-mile ride.

Rules That Never Break

Properties of Real Numbers

15 marks | 25 minutes | Part A: 10 × 1 mark · Part B: 5 marks

Name _____ Class _____ Score _____ / 15

PART A · CHOOSE THE CORRECT ANSWER

1 Which property is shown? If $13 + 25 = 38$, then $38 = 13 + 25$.

A Reflexive
B Commutative
C Symmetric
D Transitive

2 If $43 + 9 = 10 + 42$ and $10 + 42 = 52$, then $43 + 9 = \ldots$

A 52
B $10 + 42$
C 43
D $9 + 43$

3 Which property is shown by $\frac{5}{6} \cdot \frac{6}{5} = 1$?

A Commutative
B Multiplicative Inverse
C Multiplicative Identity
D Additive Inverse

4 $114 + 71 + 19 + 26 = 114 + (71 + 19) + 26$ uses the...

A Commutative Property
B Transitive Property
C Distributive Property
D Associative Property

5 Josefina wants to evaluate $\frac{7}{3} \cdot 2 \cdot 3$. What should she do first?

A Add $2 + 3$
B Divide 3 by 7
C Multiply $\frac{7}{3} \cdot 2$
D Switch 2 and 3: $\frac{7}{3} \cdot 3 \cdot 2$

6 Which property of equality justifies the statement?
$3m - 1 = 3m - 1$

A Transitive Property
B Reflexive Property
C Symmetric Property
D Substitution Property

7 $9y - 20 = ?$ (Reflexive Property)

A $9y - 20$
B $20 - 9y$
C 0
D $9y$

8 Name the property shown: $\frac{6}{2} \cdot \frac{2}{6} = 1$

A Multiplicative Inverse
B Additive Inverse
C Multiplicative Identity
D Multiplicative Property of Zero

9 Evaluate $[8 - 3(2) + 2] \cdot 0 + \frac{2}{3} \cdot \frac{3}{2}$.

A 0
B 2
C 3
D 1

10 Use the Commutative and Associative Properties to find $10 + 28 + 10 + 12$ mentally.

A 62
B 60
C 61
D 59

PART B · SHOW YOUR WORK

11 Evaluate $\frac{5}{9} \cdot 9 \cdot \frac{9}{5} \cdot 4$. Name the property used in each step.

Share It Out

Distributive Property

15 marks | 25 minutes | Part A: 10 × 1 mark · Part B: 5 marks

Name _____ Class _____ Score _____ / 15

PART A · CHOOSE THE CORRECT ANSWER

1 Simplify $4(5x - 7)$.

A $20x - 28$ B $20x + 28$
C $9x - 11$ D $20x - 7$

2 Simplify $(3y^2 + y - 8)6$.

A $18y^2 + y - 8$ B $18y^2 + 6y - 48$
C $18y^2 + 6y - 8$ D $18y^3 + 6y - 48$

3 Simplify $3(2x + 3y) - 2(4x - y)$.

A $-2x + 11y$ B $-2x + 7y$
C $-2x + 9y$ D $14x + 11y$

4 Use the Distributive Property to find $5 \cdot 104$.

A 519 B 520
C 521 D 522

5 A team practises 4 days a week: 20 min warm-up, 50 min drills and 15 min game play each day. How many minutes is that each week?

A 339 B 340
C 341 D 342

6 Simplify $-5(6m - 8)$.

A $-30m + 40$ B $m + 40$
C $-30m - 8$ D $-30m - 40$

7 Simplify $(4p^2 - 3p + 3)2$.

A $8p^2 - 3p + 3$ B $8p^2 - 6p + 3$
C $8p^3 - 6p + 6$ D $8p^2 - 6p + 6$

8 Simplify $6n + 15n$.

A $-9n$ B $90n$
C $21n$ D $21n^2$

9 Simplify $5(x + 5y) - 3(2x - 5y)$.

A $11x + 40y$ B $-x + 40y$
C $-x + 10y$ D $3x + 30y$

10 Use the Distributive Property to find $8 \cdot 99$.

A 791 B 793
C 792 D 794

PART B · SHOW YOUR WORK

11 Write an expression for "three times the sum of $2x$ and $3y$ decreased by twice the difference of $4x$ and y". Then simplify it and name the property in each step.

Say It in Math

Writing and Interpreting Equations

15 marks | 25 minutes | Part A: 10 × 1 mark · Part B: 5 marks

Name _____ Class _____ Score _____ / 15

PART A · CHOOSE THE CORRECT ANSWER

1 Which equation represents *four times a number less 10 is equal to 16*?

A $4x - 10 = 16$ B $10 - 4x = 16$
C $4(x - 10) = 16$ D $4x + 10 = 16$

2 Etu has read 12 of the 32 chapters. He plans to read **c** chapters each day for 8 days. Which equation fits?

A $12 - \frac{c}{8} = 32$ B $12 + \frac{c}{8} = 32$
C $12 - 8c = 32$ D $12 + 8c = 32$

3 Which formula gives the horsepower: the product of the speed **M** and the torque **T** divided by 5252?

A $H = \frac{M}{5252T}$ B $H = \frac{MT}{5252}$
C $H = \frac{5252}{MT}$ D $H = \frac{5252M}{T}$

4 Which sentence represents $(y + 3)^2 = 25$?

A y squared plus 3 is 25. B y plus 3 is 25.
C The quantity y plus 3 squared is 25. D y plus 3 squared is 25.

5 Write an equation for the sentence: *three times the quantity x plus six equals twenty-three.*

A $3x = 6 + 23$ B $3x + 6 = 23$
C $x + 3 \cdot 6 = 23$ D $3(x + 6) = 23$

6 Of 700 students surveyed, 294 play football, 101 play basketball, and the rest, **n**, play no sport. Which equation can find **n**?

A $294 + n = 700 + 101$ B $(294 + 101) + n = 700$
C $(294 + 101) - n = 700$ D $n = 294 + 101 + 700$

7 Translate: *The horsepower **H** of a motor is the product of the speed **M** and the torque **T** divided by 5252.*

A $H = \frac{MT}{5252}$ B $H = \frac{5252M}{T}$
C $H = \frac{5252}{MT}$ D $H = \frac{M}{5252T}$

8 Which sentence represents $(y + 2)^2 = 17$?

A y plus two squared is 17 B the quantity y plus two, squared, is 17
C y squared plus two is 17 D y plus two is 17

9 The surface area of a rectangular prism is $S = 2\ell w + 2\ell h + 2wh$. What does $2\ell h$ represent?

A the volume of the prism B the total area of the left and right faces
C the total area of the front and back faces D the area of the top face only

10 Write an equation for the sentence: *the sum of five times a number and four is fifteen.*

A $5 + 4n = 15$ B $5(n + 4) = 15$
C $5n = 4 + 15$ D $5n + 4 = 15$

PART B · SHOW YOUR WORK

11 There were 136 million packages of plain bagels sold. This is 24 million more than twice the number of raisin bagels **c**. (a) Write an equation. (b) Solve it.

One Move Wins

Solving One-Step Equations

15 marks | 25 minutes | Part A: 10 × 1 mark · Part B: 5 marks

Name _____ Class _____ Score _____ / 15

PART A · CHOOSE THE CORRECT ANSWER

1 Solve $\frac{2}{3} + w = 1\frac{1}{2}$ and name the property used.

A $\frac{5}{6}$; Addition Property of Equality
B $\frac{13}{6}$; Addition Property of Equality
C $\frac{13}{6}$; Subtraction Property of Equality
D $\frac{5}{6}$; Subtraction Property of Equality

2 A male bulldog weighs 15 lb less than a golden retriever. The bulldog weighs 50 lb. Which equation and answer are correct?

A **50 = 15 − w**; 65 lb
B **50 = w − 15**; 35 lb
C **50 = w + 15**; 35 lb
D **50 = w − 15**; 65 lb

3 Imani spent AED 146.58 on 21 yards of fabric. Which equation and price per yard are correct?

A **146.58(21) = p**; AED 3078.18
B **21p = 146.58**; AED 3078.18
C **146.58p = 21**; AED 0.14
D **21p = 146.58**; AED 6.98

4 Solve **5 + m = −4**.

A **−10**
B **−8**
C **−9**
D **−7**

5 Solve $\frac{p}{3} = 2$.

A **8**
B **7**
C **5**
D **6**

6 Player A has won 3 more titles than Player B. Player A has 22 titles. Write and solve an equation to find Player B's titles.

A **18**
B **19**
C **20**
D **21**

7 Solve **g − 5 = 100**.

A **106**
B **107**
C **105**
D **104**

8 Solve **10 = −2n**.

A **−4**
B **−6**
C **−5**
D **−3**

9 Player A has won 4 more titles than Player B. Player A has 17 titles. Write and solve an equation to find Player B's titles.

A **13**
B **14**
C **15**
D **12**

10 Solve **a − 30 = −63**.

A **−31**
B **−34**
C **−33**
D **−32**

PART B · SHOW YOUR WORK

11 96 sophomores, which is two-thirds of the class, are going to a game. (a) Write an equation. (b) Find the number of sophomores. (c) Explain why the answer is reasonable.

Undo, Step by Step

Solving Multi-Step Equations

15 marks | 25 minutes | Part A: 10 × 1 mark · Part B: 5 marks

Name _____ Class _____ Score _____ / 15

PART A · CHOOSE THE CORRECT ANSWER

1 A store sold $\frac{2}{3}$ of its basketballs, but 8 were returned. Now it has 38. Which equation and answer are correct?

A $\frac{2}{3}b + 8 = 38$; 45 B $\frac{1}{3}b - 8 = 38$; 138

C $\frac{1}{3}b + 8 = 38$; 90 D $\frac{2}{3}b - 8 = 38$; 69

2 Solve $2 - ax = -8$ for x. Assume $a \neq 0$.

A $x = \frac{6}{a}$ B $x = -\frac{6}{a}$

C $x = \frac{10}{a}$ D $x = -\frac{10}{a}$

3 Solve $-3x + 12 = -15$.

A 9 B 8

C 10 D 11

4 Solve $\frac{n+9}{7} = 4$.

A 19 B 20

C 21 D 18

5 A store sold $\frac{1}{3}$ of its basketballs, then 10 were returned. Now it has 72 basketballs. How many did it have at first?

A 95 B 94

C 92 D 93

6 Solve $12 - ax = 6$ for x. Assume $a \neq 0$.

A $x = \frac{18}{a}$ B $x = \frac{6}{a}$

C $x = 6a$ D $x = -\frac{6}{a}$

7 Solve $-5x + 17 = 12$.

A 0 B 2

C 3 D 1

8 Solve $\frac{x-6}{-3} = 9$.

A −21 B −19

C −22 D −20

9 A class raised $\frac{1}{3}$ of the cost of a trip with a bake sale, then AED 140 more from a raffle. In total they raised AED 254. What is the cost of the trip?

A 343 B 341

C 342 D 344

10 Solve $2 - ax = -5$ for x. Assume $a \neq 0$.

A $x = -\frac{7}{a}$ B $x = \frac{7}{a}$

C $x = 7a$ D $x = \frac{-3}{a}$

PART B · SHOW YOUR WORK

11 The student council raised $\frac{2}{5}$ of the money for a trip with a bake sale, then AED 150 from raffle tickets. They have AED 630. (a) Write an equation. (b) Find the cost of the trip.

Both Sides Count

Solving Equations with the Variable on Each Side

15 marks | 25 minutes | Part A: 10 × 1 mark · Part B: 5 marks

Name _____ Class _____ Score _____ / 15

PART A · CHOOSE THE CORRECT ANSWER

1 Which equation is an identity?

A $8(g + 6) = 5g + 3(g + 16)$ B $3w + 2 = 7w$
C $6(y - 5) = 2(10 + 3y)$ D $5x + 5 = 3(5x - 4) - 10x$

2 $6(y - 5) = 2(10 + 3y)$ has...

A no solution B infinitely many solutions
C the solution $y = 0$ D one solution

3 Solve $-7x - 16 = 7x + 82$.

A -6 B -7
C -5 D -8

4 Solve $7(n - 3) = 5(-3 + n) + 8$.

A 6 B 7
C 8 D 9

5 Sara has saved AED 11 and adds AED 4 a day. Omar starts with AED 0 and adds AED 5 a day. After how many days will they have the same amount?

A 13 B 10
C 12 D 11

6 Does $4(6y - 1) = 24y + 5$ have one solution, no solution, or is it an identity?

A exactly two solutions B identity (all real numbers)
C one solution D no solution

7 Solve $4x + 18 = 5x + 8$.

A 9 B 10
C 12 D 11

8 Solve $5(p - 5) = -6(5 + p) + 93$.

A 7 B 10
C 9 D 8

9 Sara has saved AED 36 and adds AED 5 a day. Omar starts with AED 0 and adds AED 8 a day. After how many days will they have the same amount?

A 11 B 12
C 13 D 14

10 Does $2(6g + 6) = 12g + 3$ have one solution, no solution, or is it an identity?

A identity (all real numbers) B one solution
C no solution D exactly two solutions

PART B · SHOW YOUR WORK

11 A rectangle has width 5 and length $x + 4$. A triangle has base 12 and height $2x - 6$. (a) Write an equation for equal areas. (b) Solve it.

Two Ways Home

Solving Equations Involving Absolute Value

15 marks | 25 minutes | Part A: 10 × 1 mark · Part B: 5 marks

Name _____ Class _____ Score _____ / 15

PART A · CHOOSE THE CORRECT ANSWER

1 Solve $|y + 2| = 4$.

A $\{-2, 6\}$ B $\{-6, 2\}$
C $\{-6, -2\}$ D $\{2\}$

2 Solve $|3x - 4| = -1$.

A $\{1\}$ B all real numbers
C $\varnothing$ D $\{1, \frac{5}{3}\}$

3 For $|ax + b| = c$, which statement must be true for the solution to be $\varnothing$?

A a is negative B b is negative
C c is positive D c is negative

4 Which equation matches a graph with points at −10 and 2?

A $|x - 4| = 6$ B $|x + 4| = 6$
C $|x + 6| = 4$ D $|x - 1| = 6$

5 Graph the solution set of $|2t - 4| = 8$. Which points are plotted?

A −2 and 2 B 2 and 6
C −6 and 2 D −2 and 6

6 Solve $|3t + 9| = 3$.

A $\{-2\}$ B $\{-4, -2\}$
C $\{-2, 4\}$ D $\{2, 4\}$

7 Solve $5|n - 3| + 7 = 17$.

A $\varnothing$ B $\{5\}$
C $\{-5, -1\}$ D $\{1, 5\}$

8 Solve $|4x - 3| = -7$.

A $\varnothing$ (no solution) B $\{-1\}$
C $\{-7, 7\}$ D all real numbers

9 A value is on average 180, give or take 15. Write and solve an equation for the maximum and minimum.

A $|x - 180| = 15$; 180 only B $|x - 180| = 15$; 165 and 195
C $|x - 15| = 180$; 165 and 195 D $|x + 180| = 15$; 165 and 195

10 Solve $|t - 9| = 10$.

A $\{-19, 1\}$ B $\{19\}$
C $\{1, 19\}$ D $\{-1, 19\}$

PART B · SHOW YOUR WORK

11 A phone holds an average of 2000 songs, give or take 250 songs. (a) Write an absolute value equation. (b) Find the maximum and minimum number of songs.

Keep the Ratio

Solving Proportions

15 marks | 25 minutes | Part A: 10 × 1 mark · Part B: 5 marks

Name _____ Class _____ Score _____ / 15

PART A · CHOOSE THE CORRECT ANSWER

1 Solve $\frac{n-4}{8} = \frac{3}{2}$.

A 16 B 4.75
C 8 D 9.33

2 Solve $\frac{x}{12} = \frac{2x-5}{18}$.

A −3.75 B 0.83
C −10 D 10

3 Solve $\frac{a}{35} = \frac{2}{10}$.

A 8 B 9
C 6 D 7

4 Solve $\frac{p}{3} = \frac{2p+3}{5}$.

A −7 B −9
C −8 D −10

5 At a constant rate of 12 free throws in 3 minutes, how many throws in 9 minutes?

A 36 B 38
C 35 D 37

6 A trail mix must be 10% raisins. How many kilograms of raisins should be mixed with 45 kg of nuts?

A 7 B 6
C 5 D 4

7 Solve $\frac{7}{14} = \frac{28}{x}$.

A 58 B 56
C 57 D 55

8 Solve $\frac{p}{8} = \frac{p+7}{9}$.

A 57 B 56
C 58 D 55

9 At a constant rate of 12 free throws in 4 minutes, how many throws in 44 minutes?

A 134 B 131
C 132 D 133

10 A trail mix must be 10% raisins. How many kilograms of raisins should be mixed with 162 kg of nuts?

A 17 B 18
C 19 D 20

PART B · SHOW YOUR WORK

11 Parts of a city sink 140 cm every 5 years. (a) Write a proportion for the sinking in 12 years. (b) Solve it. (c) Show the answer is reasonable.

Rearrange the Rule

Using Formulas

15 marks | 25 minutes | Part A: 10 × 1 mark · Part B: 5 marks

Name _____ Class _____ Score _____ / 15

PART A · CHOOSE THE CORRECT ANSWER

1 Solve $4p - 7r = pq + 16$ for p.

A $p = \frac{16 - 7r}{4 - q}$ B $p = 16 + 7r - 4q$

C $p = \frac{16 + 7r}{4 + q}$ D $p = \frac{16 + 7r}{4 - q}$

2 A jar is a cylinder: $V = \pi r^2 h$. Which formula gives the height?

A $h = \sqrt{\frac{V}{\pi r}}$ B $h = \frac{V}{r^2}$

C $h = \pi \cdot r^2$ D $h = \frac{V}{\pi r^2}$

3 $P = 40c - 300$. Which equation gives the copies c needed for a profit P?

A $c = \frac{P}{40} + 300$ B $c = \frac{P + 40}{300}$

C $c = \frac{P + 300}{40}$ D $c = \frac{P}{40} - 300$

4 Chefs need 96 cloves of garlic. A bag has 3 heads; each head has about 8 cloves. How many bags?

A 4 B 96

C 32 D 12

5 Solve $A = \frac{h(b_1 + b_2)}{2}$ for h.

A $h = \frac{A}{2(b_1 + b_2)}$ B $h = 2A - (b_1 + b_2)$

C $h = \frac{2A}{b_1} + b_2$ D $h = \frac{2A}{b_1 + b_2}$

6 A trapezoid has area 81 m² and bases 9 m and 18 m. Find its height.

A 7 B 8

C 5 D 6

7 Convert 72 km/h to metres per second. (1 km = 1000 m, 1 h = 3600 s)

A 21 B 19

C 20 D 22

8 Solve $4a - 3b = 16$ for a.

A $a = 16 + 3b - 4$ B $a = \frac{16 - 3b}{4}$

C $a = \frac{16 + 3b}{4}$ D $a = \frac{4}{16 + 3b}$

9 Profit is $P = 36c - 100$, where c is the number of copies sold. How many copies give a profit of AED 3500?

A 102 B 100

C 101 D 99

10 Convert 108 km/h to metres per second. (1 km = 1000 m, 1 h = 3600 s)

A 32 B 29

C 30 D 31

PART B · SHOW YOUR WORK

11 A recipe needs 20 fl oz of milk per batch. Nita buys half a gallon of milk. (a) Convert half a gallon to fluid ounces (1 gal = 4 qt, 1 qt = 2 pt, 1 pt = 2 cups, 1 cup = 8 fl oz). (b) How many full batches can she make?

Draw the Line

Graphing Linear Functions

15 marks | 25 minutes | Part A: 10 × 1 mark · Part B: 5 marks

Name _____ Class _____ Score _____ / 15

PART A · CHOOSE THE CORRECT ANSWER

1 What are the intercepts of $-x + 2y = 8$?

A x-int −8, y-int −4
B x-int −8, y-int 4
C x-int 8, y-int 4
D x-int 4, y-int −8

2 The graph of $y = 5$ is...

A a horizontal line through (0, 5)
B a line with slope 5
C the point (5, 5)
D a vertical line through (5, 0)

3 Which equation is NOT a function?

A $y = x + 6$
B $y = 6x$
C $y = 6$
D $x = 6$

4 In the table for $y = 4x + 3$, what is y when $x = -3$?

A −8
B −10
C −7
D −9

5 Which describes the graph of $y = 3$?

A a line through the origin with slope 3
B a single point (3, 3)
C a horizontal line through (0, 3)
D a vertical line through (3, 0)

6 Find the intercepts of $x + 5y = 5$.

A x-intercept 5, y-intercept -1
B x-intercept 1, y-intercept 5
C x-intercept -5, y-intercept 1
D x-intercept 5, y-intercept 1

7 A bag has 50 cups of food. The dog eats $\frac{5}{3}$ cups a day, so $y + \frac{5}{3}x = 50$. Find the x-intercept. What does it mean?

A 29
B 30
C 32
D 31

8 In the table for $y = -4x - 6$, what is y when $x = 3$?

A −17
B −18
C −16
D −19

9 Which describes the graph of $y = 4$?

A a single point (4, 4)
B a line through the origin with slope 4
C a vertical line through (4, 0)
D a horizontal line through (0, 4)

10 Find the intercepts of $2x + 5y = -10$.

A x-intercept -5, y-intercept -2
B x-intercept -2, y-intercept -5
C x-intercept -5, y-intercept 2
D x-intercept 5, y-intercept -2

PART B · SHOW YOUR WORK

11 A farm has 4362 lb of peanuts. $y = -\frac{2}{3}x + 4362$ gives the pounds left after x cups of peanut butter. (a) Find both intercepts. (b) Explain what they mean.

Rise Over Run

Rate of Change and Slope

15 marks | 25 minutes | Part A: 10 × 1 mark · Part B: 5 marks

Name _____ Class _____ Score _____ / 15

PART A · CHOOSE THE CORRECT ANSWER

1 The slope of the line through (2, 5) and (2, −1) is...

A 0
B −3
C undefined
D 3

2 Which table shows a linear function?

A x: 22, 29, 36 · y: −4, −1, 1
B x: 1, 2, 4 · y: 1, 2, 3
C x: 0, 2, 4 · y: 1, 4, 9
D x: 11, 8, 5 · y: −5, −3, −1

3 Find the rate of change.

x	2	5	8
y	12	21	30

A 5
B 2
C 3
D 4

4 Is the function linear?

x	20	22	24	26
y	20	22	24	26

A No, the rate of change is not constant
B Yes, the rate of change is constant
C No, because y is negative
D Yes, because x increases by the same amount

5 Find the slope of the line through **(1, −6)** and **(−1, −4)**.

A 1
B 0
C −1
D −2

6 What kind of slope does the line through **(1, −6)** and **(5, 2)** have?

A 0 (horizontal line)
B positive
C undefined (vertical line)
D negative

7 Find the rate of change.

x	4	6	8
y	-13	-23	-33

A −6
B −5
C −3
D −4

8 Is the function linear?

x	16	22	28	34
y	-5	-6	-3	-1

A No, the rate of change is not constant
B Yes, because x increases by the same amount
C Yes, the rate of change is constant
D No, because y is negative

9 Find the slope of the line through **(−2, 3)** and **(−1, −2)**.

A −6
B −5
C −3
D −4

10 What kind of slope does the line through **(0, 2)** and **(5, −8)** have?

A 0 (horizontal line)
B positive
C negative
D undefined (vertical line)

PART B · SHOW YOUR WORK

11 A student council had AED 1350 in 2000 and AED 1675 in 2005. (a) Find the rate of change. (b) Explain its meaning.

Start and Steer

Slope-Intercept Form

15 marks | 25 minutes | Part A: 10 × 1 mark · Part B: 5 marks

Name _____ Class _____ Score _____ / 15

PART A · CHOOSE THE CORRECT ANSWER

1 Which is the equation of the line with slope −5 and y-intercept 12?

A $y = -5x + 12$ B $y = 5x + 12$
C $y = -5x - 12$ D $y = 12x - 5$

2 What is the slope-intercept form of $-16x - 4y = -56$?

A $y = 16x + 56$ B $y = -4x - 14$
C $y = -4x + 14$ D $y = 4x - 14$

3 Write $-22x + 8y = 4$ in slope-intercept form.

A $y = 2.75x + 4$ B $y = 2.75x + 0.5$
C $y = 22x + 4$ D $y = -2.75x + 0.5$

4 183 million users in 2012, increasing by 9 million per year. Which equation fits (x = years after 2012)?

A $y = 9x + 183$ B $y = 192x$
C $y = 183x + 9$ D $y = 9x - 183$

5 Write the equation of the line with slope 4 and y-intercept -4.

A $y = -4x - 4$ B $y = 4x + 4$
C $y = 4x$ D $y = 4x - 4$

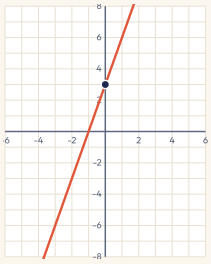
6 Write $4x + 4y = 24$ in slope-intercept form.

A $y = -x - 6$ B $y = -4x + 24$
C $y = x + 6$ D $y = -x + 6$

7 A club had 165 members in 2020 and gains about 15 members each year. Which equation gives the members y after x years?

A $y = 15x - 165$ B $y = 180x$
C $y = 15x + 165$ D $y = 165x + 15$

8 Which equation matches the graph?



A $y = 3x - 3$ B $y = -3x + 3$
C $y = 3x + 4$ D $y = 3x + 3$

9 Write the equation of the line with slope $-\frac{6}{5}$ and y-intercept -3.

A $y = -\frac{6}{5}x$ B $y = \frac{6}{5}x - 3$
C $y = -\frac{6}{5}x + 3$ D $y = -\frac{6}{5}x - 3$

10 Write $-5x + 4y = 8$ in slope-intercept form.

A $y = \frac{5}{4}x + 2$ B $y = 5x + 8$
C $y = -\frac{5}{4}x + 2$ D $y = \frac{5}{4}x - 2$

PART B · SHOW YOUR WORK

11 There were 4.61 million job openings in May; the number increased by 0.06 million per month. (a) Write an equation. (b) Explain what $x = 2$ means and find y.

Slide, Stretch, Flip

Transformations of Linear Functions

15 marks | 25 minutes | Part A: 10 × 1 mark · Part B: 5 marks

Name _____ Class _____ Score _____ / 15

PART A · CHOOSE THE CORRECT ANSWER

1 Describe **g(x) = x − 2** compared with **f(x) = x**.

A 2 units left
B 2 units down
C 2 units right
D 2 units up

2 Describe **g(x) = (x + 5)** compared with **f(x) = x**.

A 5 units up
B 5 units right
C 5 units down
D 5 units left

3 Describe **g(x) = (x − 6) + 3**.

A 3 right, 6 up
B 6 left, 3 up
C 6 right, 3 up
D 6 right, 3 down

4 The graph of **g(x) = 6x** is a ___ of the parent function and is ___.

A vertical stretch; steeper
B translation; parallel
C vertical compression; less steep
D horizontal stretch; less steep

5 Paint costs **f(p) = 6.99p**. A AED 5.95 coupon gives **g(p) = 6.99p − 5.95**. Describe **g**.

A 5.95 units up
B 5.95 units right
C 5.95 units left
D 5.95 units down

6 Describe **g(x) = (x − 6) + 9** compared with **f(x) = x**.

A translated 6 units left and 9 units up
B translated 6 units right and 9 units up
C translated 6 units right and 9 units down
D translated 9 units right and 6 units up

7 Tickets cost **f(t) = 6t**. A service fee of AED 6 is added: **g(t) = 6t + 6**. Describe **g** compared with **f**.

A translated 6 units up
B stretched by 6
C translated 6 units down
D translated 6 units left

8 Describe **g(x) = 4x** written as **g(x) = 4 · f(x)**, where **f(x) = x**.

A reflection across the y-axis
B translation 4 units up
C vertical compression (less steep)
D vertical stretch (steeper)

9 Describe **g(x) = f(−x)** where **f(x) = x + 5**.

A reflected across the x-axis
B reflected across the y-axis
C translated 5 units left
D no change

10 Describe **g(x) = x + 7** compared with **f(x) = x**.

A stretched vertically by 7
B translated 7 units down
C translated 7 units up
D translated 7 units left

PART B · SHOW YOUR WORK

11 Describe each transformation of **f(x) = x**: (a) **g(x) = (x + 2) − 4** (b) **h(x) = −2x** (c) **k(x) = f($\frac{1}{3}$x)**.

Same Step, Every Step

Arithmetic Sequences

15 marks | 25 minutes | Part A: 10 × 1 mark · Part B: 5 marks

Name _____ Class _____ Score _____ / 15

PART A · CHOOSE THE CORRECT ANSWER

1 Which is an arithmetic sequence?

A 2, 4, 8, 16, ...
B 82, 73, 64, 55, ...
C 17, 14, 10, 7, ...
D 1, 4, 9, 16, ...

2 The next three terms of 31, 18, 5, ... are...

A −8, −18, −31
B −8, −21, −34
C 8, −5, −18
D −13, −26, −39

3 Which equation gives the nth term of −4, −1, 2, 5, ...?

A $a_n = 3n + 7$
B $a_n = -4n + 3$
C $a_n = 3n - 4$
D $a_n = 3n - 7$

4 A balance is 525, 580, 635, 690, ... (months 0, 1, 2, 3). Which function fits?

A **f(n) = 580n − 55**
B **f(n) = 525n + 55**
C **f(n) = 55n + 525**
D **f(n) = 55n + 580**

5 Is **11, 13, 15, 18, 19, ...** an arithmetic sequence?

A Yes, the common difference is -2
B No, the differences are not the same
C No, because some terms are negative
D Yes, the common difference is 2

6 Find the next term of **36, 39, 42, 45, ...**

A **47**
B **48**
C **49**
D **50**

7 Write an equation for the nth term of **4, −1, −6, ...**

A $a_n = 4n - 5$
B $a_n = -5n - 9$
C $a_n = -5n + 4$
D $a_n = -5n + 9$

8 Find the 18th term of **6, 10, 14, ...**

A **74**
B **75**
C **73**
D **76**

9 An account starts with AED 400. The same amount, AED 45, is added every month. Write **f(n)** for the balance after **n** months and find it after 15 months.

A **1075**
B **1077**
C **1074**
D **1076**

10 Is **3, 1, −1, −1, −5, ...** an arithmetic sequence?

A No, because some terms are negative
B Yes, the common difference is 2
C No, the differences are not the same
D Yes, the common difference is -2

PART B · SHOW YOUR WORK

11 Randi runs mile 1 in 10.5 min, 2 miles in 21 min, 3 miles in 31.5 min and 4 miles in 42 min. (a) Write a function for her time. (b) How long for a 26.2-mile marathon?

Piece by Piece

Piecewise and Step Functions

15 marks | 25 minutes | Part A: 10 × 1 mark · Part B: 5 marks

Name _____ Class _____ Score _____ / 15

PART A · CHOOSE THE CORRECT ANSWER

1 The range of $f(x) = 2x + 4$ $(x \le 1)$, $-x + 3$ $(x > 1)$ is...

A $y \ge 6$ B all real numbers
C $y \le 6$ D $y \le 2$

2 The range of $f(x) = [\![x]\!]$ is...

A $\{0, 1\}$ B all real numbers
C all integers D $y \ge 0$

3 $f(x) = \{ -2x - 1$ if $x \le 1$; $x - 3$ if $x > 1$. Find $f(1)$.

A -2 B -1
C -3 D -4

4 Evaluate $[\![-5]\!]$ (greatest integer).

A -5 B -6
C -3 D -4

5 Boarding a dog costs AED 40 per day, and every part of a day counts as a full day. What is the cost for 3 days?

A 119 B 120
C 122 D 121

6 One piece of a function is $y = x$ for $x \le -2$. What is the range of this piece?

A $y \ge -2$ B $y \le -2$
C all real numbers D $y < -2$

7 $f(x) = \{ -x + 5$ if $x \le -3$; $3x + 3$ if $x > -3$. Find $f(-6)$.

A 13 B 11
C 10 D 12

8 For $f(x) = [\![x - 2]\!]$, find $f(-2.8)$.

A -6 B -4
C -5 D -3

9 Boarding a dog costs AED 25 per day, and every part of a day counts as a full day. What is the cost for 6 days?

A 151 B 150
C 149 D 152

10 One piece of a function is $y = -x - 3$ for $x > 1$. Which point is drawn with an open circle?

A $(1, -2)$ B $(0, -3)$
C $(2, -5)$ D $(1, -4)$

PART B · SHOW YOUR WORK

11 Boarding a dog costs AED 35 per day; every part of a day counts as a full day. (a) Make a table for $0 < x \le 4$ days. (b) Find the cost for 3.2 days.

The V Shape

Absolute Value Functions

15 marks | 25 minutes | Part A: 10 × 1 mark · Part B: 5 marks

Name _____ Class _____ Score _____ / 15

PART A · CHOOSE THE CORRECT ANSWER

1 Describe $g(x) = |x| - 3$ compared with $f(x) = |x|$.

A 3 units up
B 3 units right
C 3 units left
D 3 units down

2 The vertex of $g(x) = |x - 2| + 3$ is...

A (3, 2)
B (2, −3)
C (−2, 3)
D (2, 3)

3 Which describes $p(x) = |2x|$?

A vertical compression
B horizontal stretch
C translation 2 right
D horizontal compression

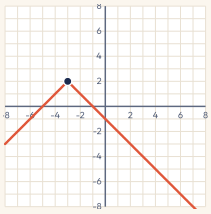
4 A graph is the parent graph moved 2 units left and 5 units down. Its equation is...

A $g(x) = |x + 2| + 5$
B $g(x) = |x + 2| - 5$
C $g(x) = |x - 5| + 2$
D $g(x) = |x - 2| - 5$

5 Describe $g(x) = |x + 6|$ compared with $f(x) = |x|$.

A reflected across the x-axis
B translated 6 units down
C translated 6 units left
D translated 6 units right

6 Which equation matches the graph?



A $g(x) = -|x + 3| + 2$
B $g(x) = -|x - 3| + 2$
C $g(x) = |x + 3| + 2$
D $g(x) = -|x + 3| - 2$

7 Which describes $g(x) = |\frac{1}{2}x|$ compared with $f(x) = |x|$?

A horizontal stretch
B horizontal compression
C vertical stretch
D vertical compression

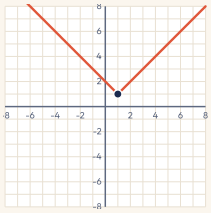
8 For $g(x) = -|x + 4| - 4$, find $g(6)$.

A −12
B −13
C −15
D −14

9 Describe $g(x) = |x + 4|$ compared with $f(x) = |x|$.

A translated 4 units left
B reflected across the x-axis
C translated 4 units down
D translated 4 units right

10 Which equation matches the graph?



A $g(x) = |x - 1| + 1$
B $g(x) = |x - 1| - 1$
C $g(x) = |x + 1| + 1$
D $g(x) = -|x - 1| + 1$

PART B · SHOW YOUR WORK

11 For $j(x) = -|x + 3| + 5$: (a) describe the transformations of $f(x) = |x|$, (b) give the vertex, (c) state the range.